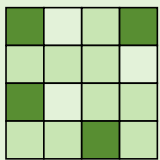


Input I 

Structure-aware Mamba Encoder

Conv
Embedding
($k=7, s=4$)

Stage1

SAME
Block
 $\times L_1$ Downsample
 $\times 2$

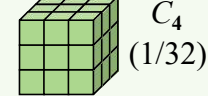
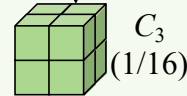
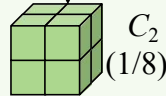
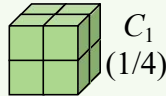
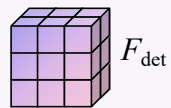
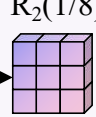
Stage2

SAME
Block
 $\times L_2$ Downsample
 $\times 2$

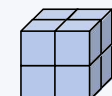
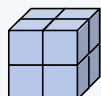
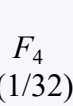
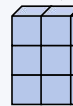
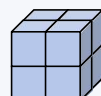
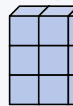
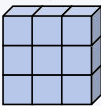
Stage3

SAME
Block
 $\times L_3$ Downsample
 $\times 2$

Stage4

SAME
Block
 $\times L_4$ High-resolution
detail Branch I Conv 3×3 Conv 3×3  $R_1(1/4)$  $R_2(1/8)$  $R_3(1/16)$ 

Detail-aware Hierarchical Decoder

DHD
Block₄DHD
Block₃DHD
Block₂DHD
Block₁

Output Refinement & Prediction

Channel
RecalibrationGAP $\rightarrow$ MLP $\rightarrow$
Sigmod $\rightarrow \otimes$ Up $\times 4$ $U_{1 \rightarrow 0}(F_c)$

Concat

Conv 3×3 + GN + ReLU

Segmentation Head

Prediction

